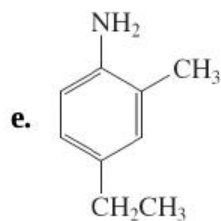
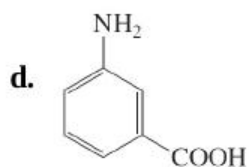
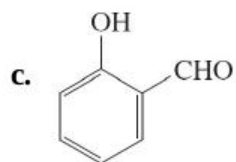
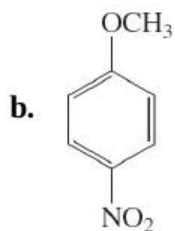
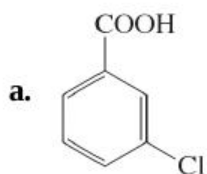
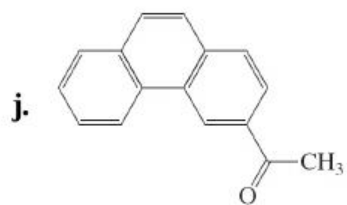
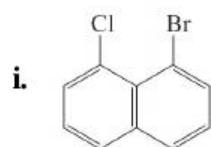
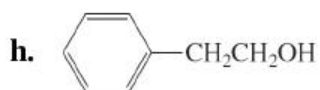
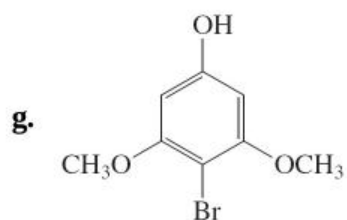
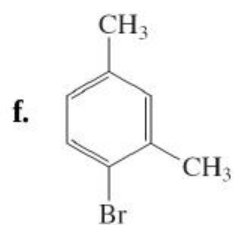


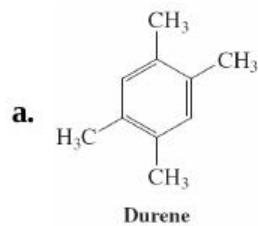
Problems

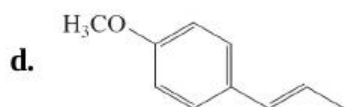
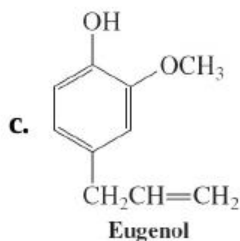
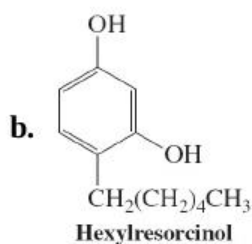
36. Name each of the following compounds by using the IUPAC system and, if possible, a reasonable common alternative. (**Hint:** The order of functional group precedence is $-\text{COOH} > -\text{CHO} > -\text{OH} > -\text{NH}_2$. $-\text{COOH} > -\text{CHO} > -\text{OH} > -\text{NH}_2$.)





37. Give a proper IUPAC name for each of the following commonly named substances.

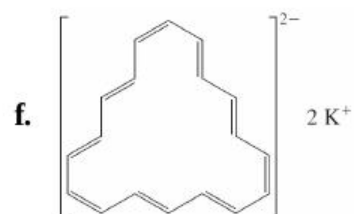
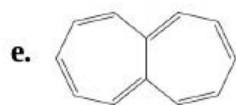
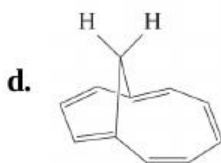
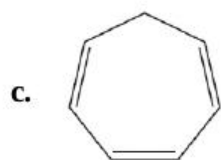
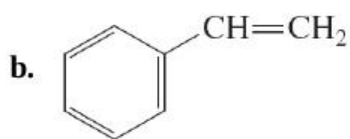


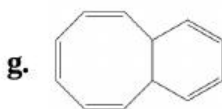


38. Draw the structure of each of the following compounds. If the name itself is incorrect, give a correct systematic alternative. (a) *o*-Chlorobenzaldehyde; (b) 2,4,6-trihydroxybenzene; (c) 4-nitro-*o*-xylene; (d) *m*-isopropylbenzoic acid; (e) 4,5-dibromoaniline; (f) *p*-methoxy-*m*-nitroacetophenone.
39. The complete combustion of benzene is exothermic by approximately $-789 \text{ kcal mol}^{-1}$. What would this value be if benzene lacked aromatic stabilization?
40. The $^1\text{H NMR}$ spectrum of naphthalene shows two multiplets (Figure 15-16). The upfield absorption ($\delta = 7.49 \text{ ppm}$) is due to the hydrogens at C2, C3, C6, and C7, and the downfield multiplet ($\delta = 7.86 \text{ ppm}$) is due to the hydrogens at C1, C4, C5, and C8. Explain why one set of hydrogens is deshielded more than the other.
41. Complete hydrogenation of 1,3,5,7-cyclooctatetraene is exothermic by

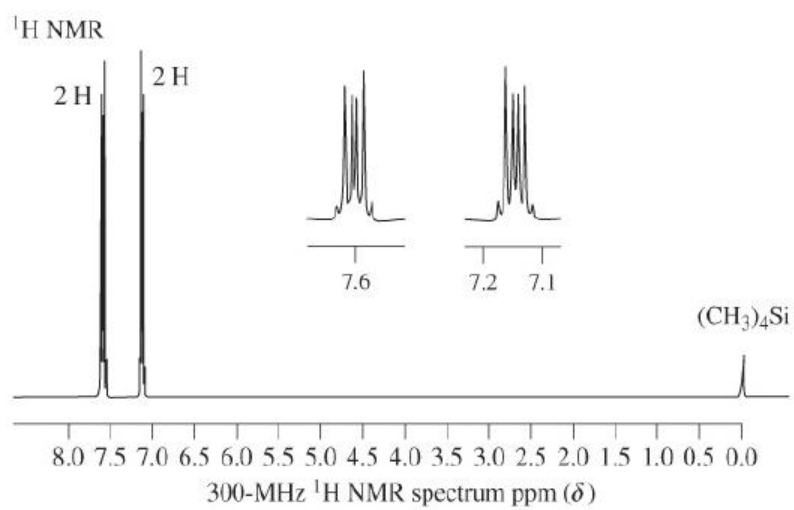
$-101 \text{ kcal mol}^{-1}$. Hydrogenation of cyclooctene proceeds with $\Delta H^\circ = -23 \text{ kcal mol}^{-1}$. Are these data roughly consistent with the description of cyclooctatetraene presented in the chapter?

42. Which of the following structures qualify as being aromatic, according to Hückel's rule?

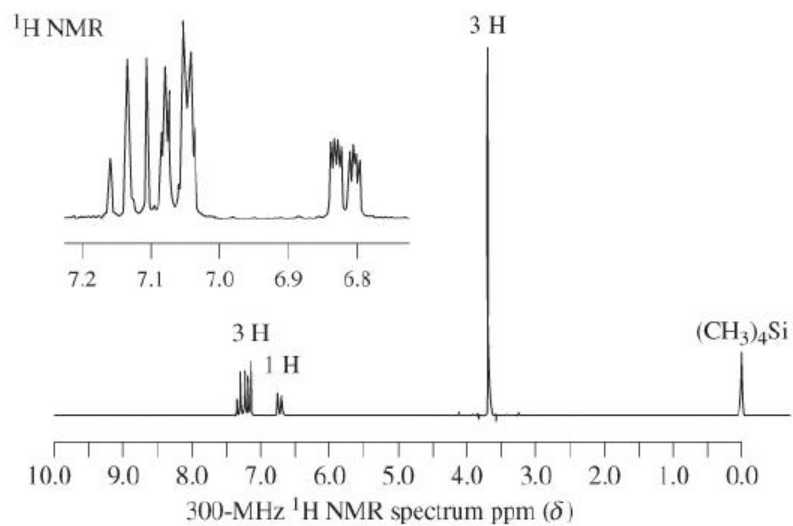




43. Following are spectroscopic and other data for several compounds. Propose a structure for each of them. **(a)** Molecular formula = $C_6H_4Br_2$.
Molecular formula = $C_6H_4Br_2$. 1H NMR spectrum A.
spectrum A. ^{13}C NMR: 3 peaks; ^{13}C NMR : 3 peaks; IR:
 $\tilde{\nu} = 745$ (s, broad) cm^{-1} ; IR : $\tilde{\nu} = 745$ (s, broad) cm^{-1} ;
UV : $\lambda_{max}(\epsilon) = 263(150)$, UV : $\lambda_{max}(\epsilon) = 263(150)$, 270(250), 270(250),
278(180) nm. 278(180) nm. **(b)** Molecular formula = C_7H_7BrO .
Molecular formula = C_7H_7BrO . 1H NMR spectrum B.
spectrum B. ^{13}C NMR: 7 peaks; ^{13}C NMR : 7 peaks;
IR : $\tilde{\nu} = 680$ (s) and 765 (s) cm^{-1} . IR : $\tilde{\nu} = 680$ (s) and 765 (s) cm^{-1} .
(c) Molecular formula = $C_9H_{11}Br$. Molecular formula = $C_9H_{11}Br$.
 1H NMR spectrum C. spectrum C.
 ^{13}C NMR (DEPT) : $\delta = 20.6(CH_3)$, $23.6(CH_3)$,
 ^{13}C NMR (DEPT) : $\delta = 20.6(CH_3)$, $23.6(CH_3)$, $124.2(C_{quat})$,
 $124.2(C_{quat})$, $129.0(CH)$, $129.0(CH)$, $136.0(C_{quat})$, $136.0(C_{quat})$,
 $137.7(C_{quat})$ ppm. $137.7(C_{quat})$ ppm.

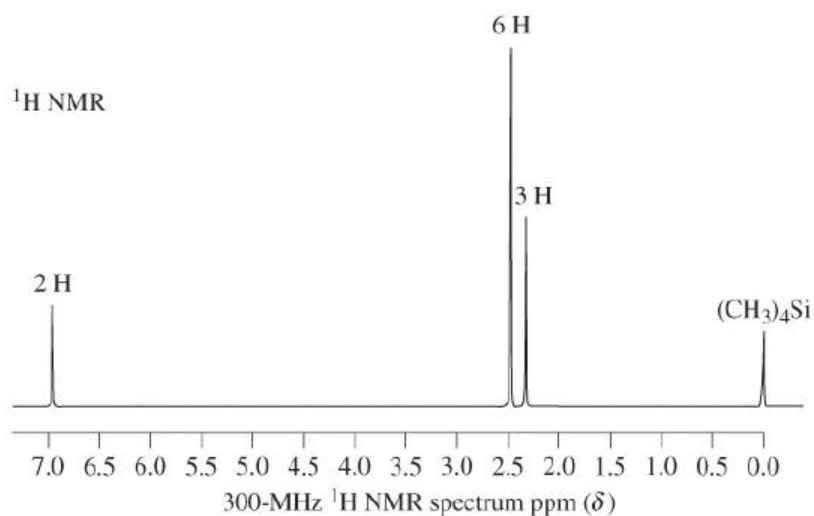


A



B

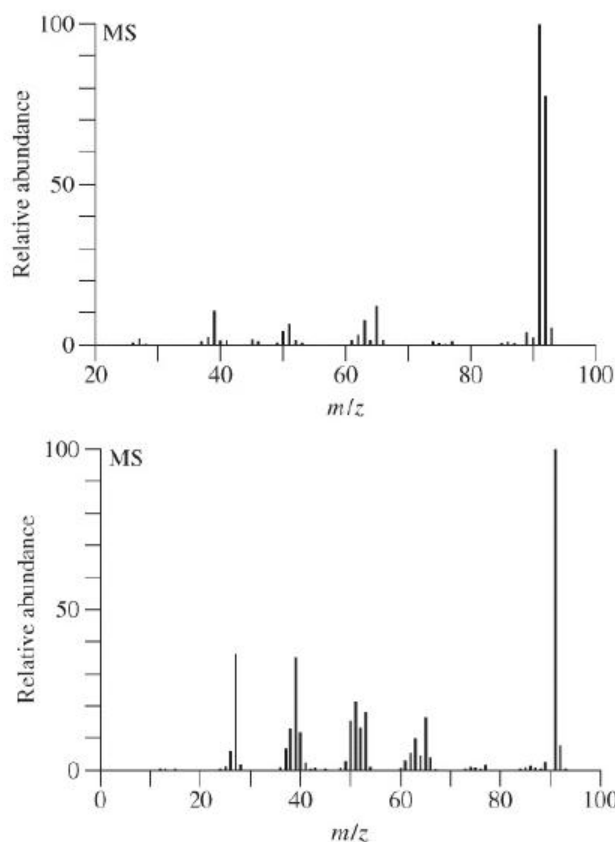
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C

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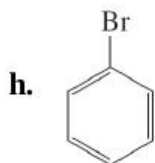
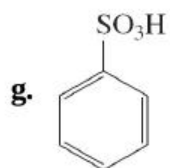
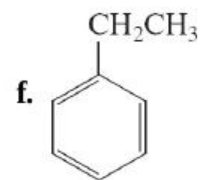
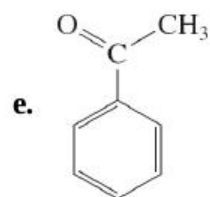
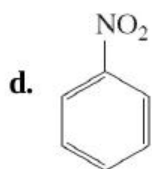
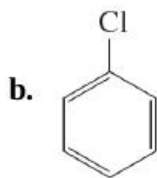
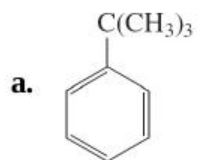
44. Both methylbenzene (toluene) and 1,6-heptadiyne have molecular formulas of C_7H_8 and molecular masses of 92. Which of the two mass spectra shown below corresponds to which compound? Explain your reasoning.

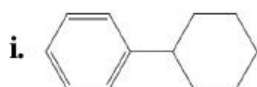


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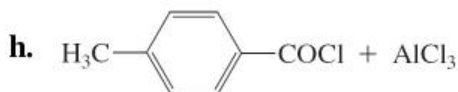
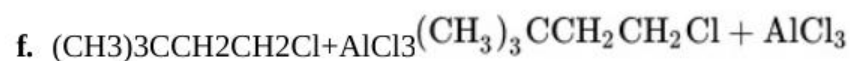
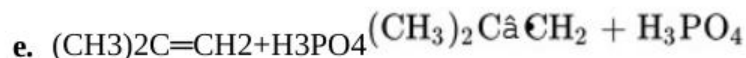
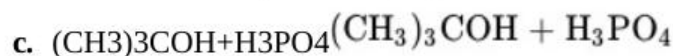
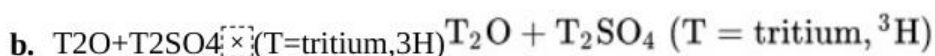
45. (a) Is it possible to distinguish the three isomers of dimethoxybenzene solely on the basis of the number of peaks in their proton-decoupled ^{13}C NMR spectra? Explain. (b) How many different isomers of dimethoxynaphthalene exist? How many peaks should each one exhibit in the proton-decoupled ^{13}C NMR spectrum?
46. The species resulting from the addition of benzene to HF-SbF_5 ([Exercise 15-24](#)) shows the following C NMR absorptions: ^{13}C NMR absorptions: $\delta = 52.2(\text{CH}_2)$, ^{13}C NMR absorptions : $\delta = 52.2(\text{CH}_2)$, $136.9(\text{CH})$, $136.9(\text{CH})$, $178.1(\text{CH})$, $178.1(\text{CH})$, and $186.6(\text{CH})$ ppm. The signals at $\delta = 136.9$ and $\delta = 186.6$ are twice the intensity of the other peaks. Assign this spectrum.
47. Reaction review. Without consulting the Reaction Road Map on [p. 749](#),

suggest a reagent to convert benzene into each of the following compounds.





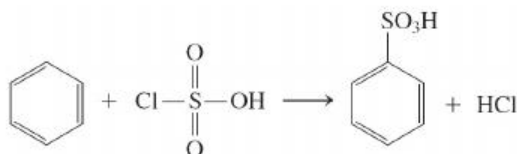
48. Give the expected major product of addition of each of the following reagent mixtures to benzene. (**Hint:** Look for analogies to the reactions presented in this chapter.)



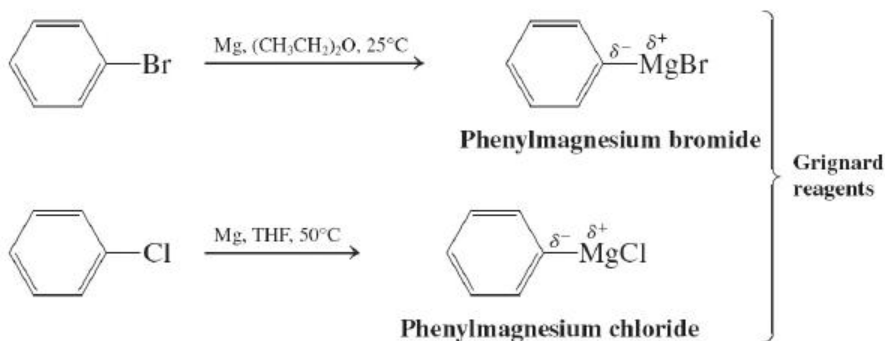
49. Write mechanisms for the reactions of parts (c) and (f) in [Problem 48](#).

50. Hexadeuteriobenzene, C_6D_6 , is a very useful solvent for ^1H NMR spectroscopy because it dissolves a wide variety of organic compounds and, being aromatic, is very stable. Suggest a method for the preparation of C_6D_6 .

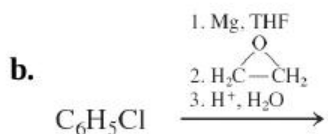
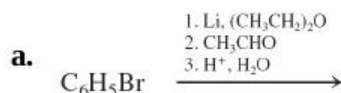
51. Propose a mechanism for the sulfonation of benzene using chlorosulfuric acid, ClSO_3H (given below).



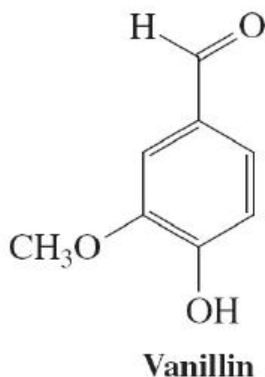
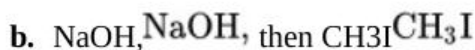
52. Benzene reacts with sulfur dichloride, SCl_2 , in the presence of AlCl_3 to give phenylthiobenzene, $\text{C}_6\text{H}_5\text{-S-C}_6\text{H}_5$. Propose a mechanism for this process.
53. (a) 3-Phenylpropanoyl chloride, $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{COCl}$, reacts with AlCl_3 to give a single product with the formula $\text{C}_9\text{H}_8\text{O}$ and an $^1\text{H NMR}$ spectrum with signals at $\delta = 2.53$ (t, $J = 8 \text{ Hz}$, 2 H), 3.02 (t, $J = 8 \text{ Hz}$, 2 H), and $7.2\text{--}7.7$ (m, 4 H) ppm. Propose a structure and a mechanism for the formation of this derivative.
- (b) The product of the process described in (a) is subjected to the following reaction sequence: (1) NaBH_4 , $\text{CH}_3\text{CH}_2\text{OH}$; (2) conc. H_2SO_4 , 100°C ; (3) H_2 , Pd-C , $\text{CH}_3\text{CH}_2\text{OH}$. The resulting molecule exhibits five resonance lines in its $^{13}\text{C NMR}$ spectrum. What is the structure of the substance formed after each of the steps in this sequence?
54. The text states that alkylated benzenes are more susceptible to electrophilic attack than is benzene itself. Draw a graph like that in Figure 15-20 to show how the energy profile of electrophilic substitution of methylbenzene (toluene) would differ quantitatively from that of benzene.
55. Like haloalkanes, haloarenes are readily converted into organometallic reagents, which are sources of nucleophilic carbon.



The chemical behavior of these reagents is very similar to that of their alkyl counterparts. Write the main product of each of the following sequences.

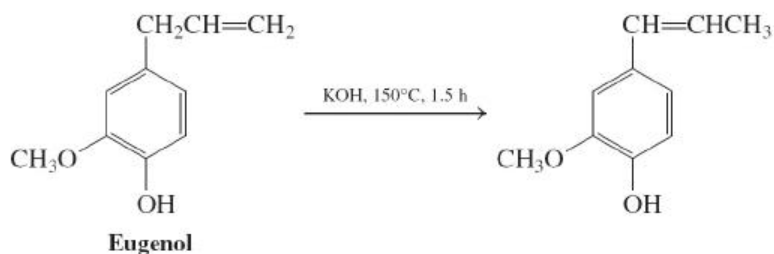


56. Give efficient syntheses of the following molecules, beginning with benzene. **(a)** 1-Phenyl-1-heptanol; **(b)** 2-phenyl-2-butanol; **(c)** octylbenzene. **(Hint:** Use a method from [Section 15-13](#). Why will Friedel-Crafts alkylation not work?)
57. Vanillin is a benzene derivative with several functional groups, each one of which displays its characteristic reactivity. What would you expect to be the products of reaction of vanillin with each of the following reagents?



Vanillin is obtained by extraction from the seed pods of plants in the *Vanilla*

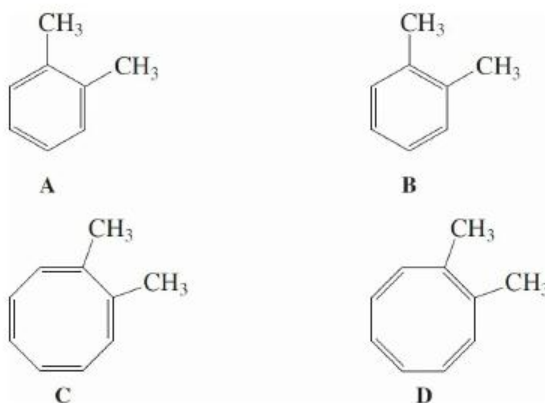
genus in a process that dates back at least 500 years: The Mexican Aztecs used it to flavor xo-coatl, a chocolate drink. Cortez discovered it in the court of Montezuma and was responsible for introducing it to Europe. Increasing demand for vanillin necessitated development of synthetic procedures, which involve extraction of related substances from other plant-derived sources. One of the most important of these sources is the wood-derived waste from the manufacture of paper. Treatment with aqueous NaOH, **NaOH**, followed by oxidation with pressurized air at 170°C **170 °C** produces sizeable quantities of vanillin. The conversion of eugenol (an extract from cloves) into vanillin is very similar chemically. First, treatment of eugenol with KOH **KOH** at 150 °C **150 °C** in a high-boiling solvent causes positional isomerization of its side-chain double bond:



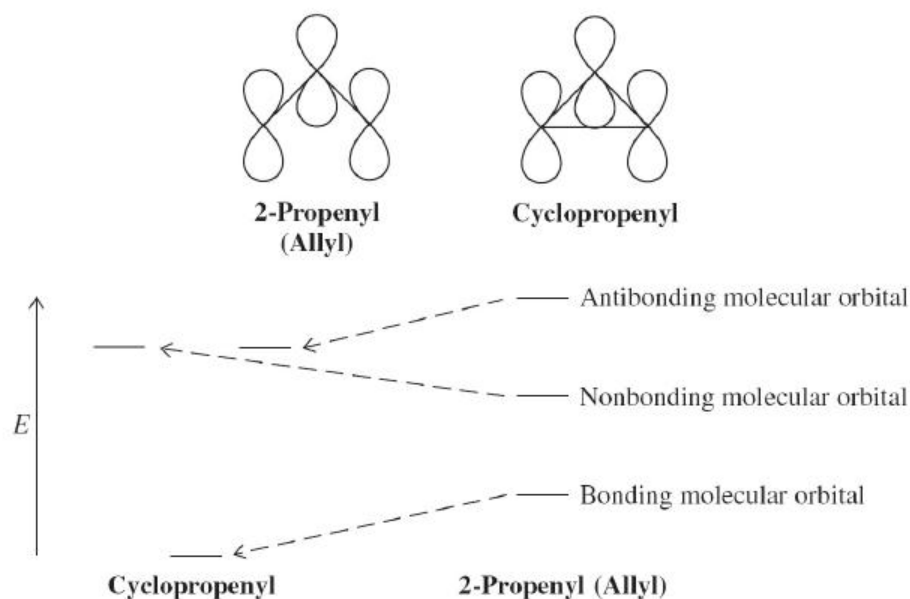
Subsequently, oxidative cleavage (see [Section 12-12](#)) completes the synthesis of vanillin.

c. Propose a mechanism for the isomerization depicted in the scheme.

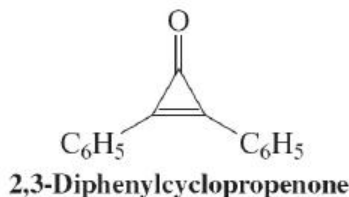
58. Because of cyclic delocalization, structures A and B shown here for *o*-dimethylbenzene (*o*-xylene) are simply two resonance forms of the same molecule. Can the same be said for the two dimethylcyclooctatetraene structures C and D? Explain.



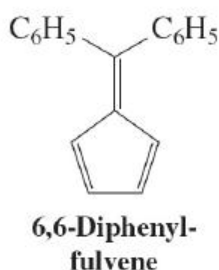
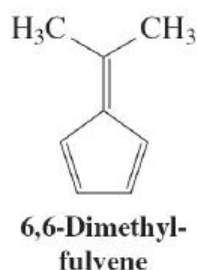
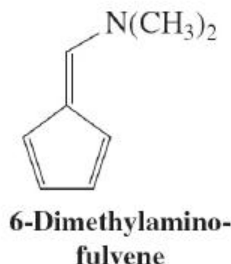
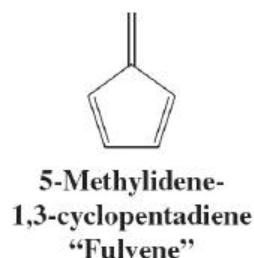
59. The energy levels of the 2-propenyl (allyl) and cyclopropenyl $\pi\pi$ systems are compared qualitatively in the diagram on the next page. **(a)** Draw the three molecular orbitals of each system, using plus and minus signs and dotted lines to indicate bonding overlap and nodes, as in [Figure 15-4](#). Does either of these systems possess degenerate molecular orbitals? **(b)** How many $\pi\pi$ electrons would give rise to the maximum stabilization of the cyclopropenyl system, relative to 2-propenyl (allyl)? (Compare [Figure 15-5](#), for benzene.) Draw Lewis structures for both systems with this number of π electrons and any appropriate atomic charges. **(c)** Could the cyclopropenyl system drawn in (b) qualify as being “aromatic”? Explain.



60. **CHALLENGE** 2,3-Diphenylcyclopropenone (see structure below) forms an addition product with HBr that exhibits the properties of an ionic substance (i.e., a salt). Suggest a structure for this product and a reason for its existence as a stable entity.

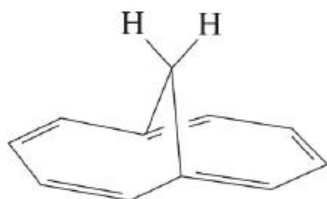


61. **CHALLENGE** Is cyclobutadiene dication ($C_4H_4^{2+}$) aromatic according to Hückel's rule? Sketch its $\pi\pi$ molecular orbital diagram to illustrate your answer.
62. All the molecules shown below are examples of "fulvenes," or methylenecyclopentadienes.



- a. One of these structures is considerably more acidic than the others, with a pK_a around 20. Identify it and its most acidic hydrogen(s), and explain why it is an unusually strong acid for a molecule with only carbon-hydrogen bonds.
- b. None of the 7-methyldiene-1,3,5-cycloheptatrienes that correspond to the fulvene structures above show any unusual acidity. Explain.
63. A characteristic reaction of fulvenes is nucleophilic addition. To which carbon in a fulvene would you expect nucleophiles to add, and why?
64. **CHALLENGE** (a) The 1H NMR spectrum of [18]annulene shows two signals, at $\delta = 9.28$ (12 H) and -2.99 (6 H) ppm. The negative chemical shift value refers to a resonance upfield (to the right) of $(CH_3)_4Si$. Explain this spectrum. (Hint: Consult Figure 15-9.) (b) The unusual molecule 1,6-methano[10]annulene (shown below) exhibits two sets of signals in the 1H NMR spectrum at $\delta = 7.10$ (8 H)

$\delta = 7.10$ (8 H) and -0.50 (2 H) ppm. -0.50 (2 H) ppm. Is this result a sign of aromatic character?

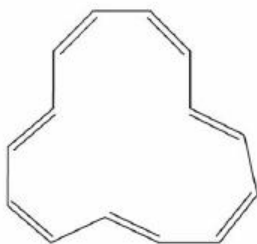


1,6-Methano[10]annulene

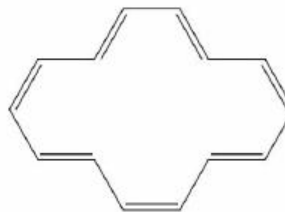
(c) The molecule in part (d) of [Problem 42](#) (1,5-methano[10]-annulene) also exhibits unusual signals in both its ^1H and ^{13}C NMR spectra: ^1H NMR: $\delta = -0.95$ (d, $J = 12$ Hz, 1 H)

^1H NMR : $\delta = -0.95$ (d, $J = 12$ Hz, 1 H) and -0.50 (d, $J = 12$ Hz, 1 H) ppm, in addition to signals between $\delta = 6.88$ and 8.11 ppm (8 H total); ^{13}C NMR: $\delta = 34.7$ ppm, in addition to signals between $\delta = 125.1$ and 161.2 ppm. Again, provide a rationale.

65. The ^1H NMR spectrum of the most stable isomer of [14]annulene shows two signals, at $\delta = -0.61$ (4 H) and 7.88 (10 H) ppm. Two possible structures for [14]annulene are shown here. How do they differ? Which one corresponds to the NMR spectrum described?



A



B

66. Explain the following reaction and the indicated stereochemical result mechanistically.